

RUTGERS-THE STATE UNIVERSITY

Master's Comprehensive Examination

PART I

THEORY

December 6, 1980

- Instructions:
1. Open Book and Open Notes.
 2. Do four out of the following six problems
 3. Use a separate blue book for this section of the exam
 4. Indicate which questions you wish graded
-

✓ 1. Let X be distributed with density function

$$\begin{aligned} f(x) &= ax && \text{for } 0 < x < 1 \\ &= bx && \text{for } 1 < x < 2 \\ &= 0 && \text{otherwise.} \end{aligned}$$

$E(X) = 1$. Find a and b .

✓ 2. Let $Y_1, Y_2, \dots, Y_5$ be a random sample from the uniform distribution defined over $(0, \theta)$. Let $X = 5 \min(Y_1, Y_2, \dots, Y_5)$. Is X an unbiased estimator of θ ? (Prove or find the bias).

3. Suppose $n = 2$ independent Bernoulli trials yield $y = 1$ success. Find the exact 95% confidence interval for p .

✓ 4. Let X be distributed with density

$$\begin{aligned} f(x) &= (2/3)x && \text{for } 1 < x < 2, \\ &= 0 && \text{otherwise.} \end{aligned}$$

Let $Y = \log_e X$. Find the moment generating function of Y .

✓ 5. Let $X_1, \dots, X_n$ be independently and identically distributed random variables each having density function

$$\begin{aligned} g(x; \theta) &= \theta x^2 \exp(-\tfrac{1}{2} \theta x^2) && \text{if } x > 0, && \theta > 0. \\ &= 0 && \text{otherwise.} \end{aligned}$$

- a) Find a (non-trivial) sufficient statistic for θ .
- b) Find the maximum likelihood estimator for θ .

6. Given a random sample of size n from the distribution with density

$$f(x) = (1 + \theta) x^\theta, \text{ for } 0 \leq x \leq 1,$$

use the Neyman-Pearson lemma to determine the nature of the best critical region for testing $H_0: \theta = 3$ versus $H_1: \theta = 2$.

RUTGERS-STATE UNIVERSITY

Master's Comprehensive Examination

PART II

Applied

December 6, 1980

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-

1. In order to characterize the effect of food consumption (X) on the gain in weight (Y) of animals on a certain diet, twenty rats were individually caged and their grams of food consumed and grams of weight gained were measured for the interval of interest. The following calculations have been made:

$$\begin{array}{lll} n = 20 & \Sigma(Y - \bar{Y})^2 = 600 & \Sigma(X - \bar{X})^2 = 90 \\ \Sigma Y = 1200 & \Sigma(X - \bar{X})(Y - \bar{Y}) = 180 & \\ \Sigma X = 400 & & \end{array}$$

- a) Estimate the correlation between food consumed and weight gained. Is it significantly different from zero?
 - b) Complete the regression analysis, testing the significance of the regression of weight gain on food consumption.
 - c) Compute the regression equation for the prediction of weight gain from food consumption.
 - d) What are the 95% confidence limits for the prediction (at the mean food consumption for this study) of weight gains from food consumption?
 - e) Construct a 95% confidence interval and make the appropriate verbal statement for the regression coefficient.
-
2. An experimenter wishes to compare six treatments. He has to choose among a number of possible designs. For the same cost he has his choice of:
- (a) 36 units which come naturally classified into 6 rows and 6 columns and for which the anticipated residual variance is 10.
 - (b) 48 units which come naturally classified into 4 blocks of 12 units each and having an anticipated within-block variance of 14.

Master's Comprehensive Examination

PART II , page 2

Applied

December 6, 1980

(2 continued):

(c) 30 units which come naturally classified into 5 blocks of 6 units each with anticipated within block variance of 8.

(d) 21 units which come naturally classified into 7 blocks of 3 units each with anticipated within block variance of 3.

Assuming that interactions can be ignored

Give your recommended design for each of these possibilities.

Give your order of preference (with justification) based on statistical considerations only.

3) A chemical procedure designed to determine the concentration of a chemical z is known from past experience, to be subject to an error variance which is 10% of the actual concentration of z in the solution.

For process control purposes it is necessary to check on the concentration of z . If the actual z concentration is 4 units then the process is running well and should be undisturbed. However, if the true z concentration rises to 5 units then it is very desirable to effect a correction.

Using whatever approximations, assumptions and judgment you feel are appropriate, give your recommendations for the number of replicate analyses to be done per sample. Be sure to give as much detail as possible.

4) A triangle test is a procedure in which a judge is presented with three samples, two that are the same and one that is different. The judge is then asked to select the odd sample. (This type of test is used quite often in the food industry). In a preliminary trial a prospective judge was successful in selecting the odd sample 11 times in 15 trials. Would you select him for the panel? How many would he have to pick correctly to be chosen? What is the probability of a Type I error if we accept the above judge for our panel?

Master's Comprehensive Examination

PART II, page 3

Applied

December 6, 1980.

5a) A study was conducted to compare two headache remedies. Two hundred patients who have a headache are randomly assigned to one of the two remedies, one hundred per group. The number of patients whose headache is cured within each group is recorded in the following table:

		Remedy		
		A	B	
Result	Cured	50	70	120
	Not Cured	50	30	80
		100	100	200

Analyze the data in this table with the objective of establishing if there is a significant difference between the remedies.

b) A study was conducted to compare two headache remedies. One hundred patients who have a headache are randomly assigned to one of the two remedies, 50 patients per remedy. The next time one of these patients has a headache, they are administered the headache remedy that they did not receive during the first episode. The results of this study are summarized in the following table:

		Remedy A		
		Cured	Not Cured	
Remedy B	Cured	35	35	70
	Not Cured	15	15	30
		50	50	100

In this table one finds that 35 patients were cured by both remedies while 15 patients were cured with Remedy A but not with Remedy B. Analyze the data in this table with the objective of establishing if there is a significant difference between the remedies.

Master's Comprehensive Examination

PART II, page 4

Applied

December 6, 1980

5(continued) c) You will note that the studies described in (a) and (b) above while having the same objective are fundamentally different in their design. Which study do you feel is "better" and why? Be as specific as possible.

6) A biologist is seeking a way of distinguishing between the organisms, A and B. She decides to seek a growing medium in which the two organisms would have sharply different growing rates. She runs the following experiment:

Day	Nutrient					
	1		2		3	
	organism		organism		organism	
	A	B	A	B	A	B
1	3	4	1	6	7	6
2	6	5	2	8	10	8
	9	9	3	14	17	14
	18		17		31	
					66	

In the above table the response is the volume of growth.

Nutrient and organism may be treated as fixed effects. The biologist is only able to prepare six Ager dishes per day so she decides to include day 1 and day 2 as a block effect.

- Construct the ANOVA table including the ems column. Show calculations.
- Relate the question asked by the biologist to the ANOVA table and to the original data.
- Does the biologist have a method that can distinguish between the organisms?
- Calculate a 95% confidence interval for the mean volume of growth for organism A

RUTGERS-THE STATE UNIVERSITY
Master's Comprehensive Examination

PART I

Theory

March 8, 1980

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1. Find the approximate probability that the number of aces obtained in 12,000 rollings of a die is between 1900 and 2150.

(Note: answer cannot be given in terms of a sum.)

2. Let $X_1, X_2, \dots, X_n$ be independent Poisson random variables with parameter λ .

Find the conditional distribution of X_1 given $S_n = \sum_{i=1}^n X_i$.

3. Let X be a normal random variable with mean μ and variance 4. Find the power of the uniformly most powerful test of the null hypothesis that $\mu = 0$ vs $\mu > 0$ at the value $\mu = 2$.

4. Find the maximum likelihood estimator of EZ for $f(z(v)) = ve^{-vz}$, $z > 0$ based on a random sample of size n . Prove whether this estimator is minimum variance unbiased.

RUTGERS-THE STATE UNIVERSITY

Master's Comprehensive Examination

PART II

Applied

March 8, 1980

- Instructions: 1. Open Book and Open Notes.
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1. a. A psychologist conducted an experiment to determine whether student's achievement is related to the teaching method used in a particular subject. Three teaching methods are involved, I, II, and III, and students' achievement is classified as A, B, or C. The results were:

Achievement	Teaching Methods			Total
	I	II	III	
A	5	20	5	30
B	15	15	10	40
C	5	15	10	30
Total	25	50	25	100

Test the null hypothesis that students' achievement is not related to the methods of teaching at $\alpha = 0.01$.

- b. Four balanced coins are tossed together 160 times. The results are shown below:

Number of Heads	0	1	2	3	4	Total
Observed Frequency	16	35	55	48	6	160

At $\alpha = 0.05$, test the null hypothesis that the four coins are all well balanced and randomly tossed.

2. Let X be the aptitude test score and Y be the hourly output of any production worker. The record of ten such workers yields the following data:

$$\begin{array}{lll} n = 10 & \Sigma X = 550 & \Sigma Y = 680 \\ \Sigma XY = 45,900 & \Sigma X^2 = 38,500 & \Sigma Y^2 = 56,000 \end{array}$$

- Determine the regression of Y on X for the above data.
- Is there sufficient evidence to conclude that the regression coefficient is different from 0 at $\alpha = 0.02$?
- Find the 98 percent confidence interval for $\mu_{Y|X}$ at $X = 40$.
- Find the 98 percent confidence interval for an individual with $X = 40$.
- Explain the differences, if any, between the answers in c and d above.

3. Suppose that a manufacturing company divides its market into three regions and classifies its salespersons into three age groups. The management wishes to determine whether there is any difference in the sales of the salespersons in different age brackets, and in the sales of the regions. Further, it wishes to find out whether there is any difference in the interaction effects between the ages of the salespersons and the regions covered. The data on sales (in thousand of dollars) for two weeks are collected and tabulated below.

Regions	Age Brackets		
	C_1	C_2	C_3
R_1	10	7	5
	12	8	3
R_2	13	9	4
	13	10	5
R_3	12	10	6
	10	8	5

It is assumed that the data are collected from a normally distributed population. Test the null hypotheses that there is no difference in sales among the age groups of salespersons, no difference in sales among the regions, and no difference in the interaction effects at $\alpha = 0.05$.

RUTGERS-THE STATE UNIVERSITY

Master's Comprehensive Examination

PART I

Theory

December 8, 1979

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1. Consider an urn containing 10 balls of which 5 are black. Choose an integer n at random from the set 1, 2, 3, and then choose a sample of size n without replacement from the urn.

(a) Find the probability that all the balls in the sample are black.

(b) Find the expected number of black balls chosen.

2. Let X, Y, Z be continuous random variables with the following density

$$f(x,y,z) = \begin{cases} 2 & \text{if } 0 < x < y < 1, 0 < z < 1 \\ 0 & \text{otherwise} \end{cases}$$

Find

(a) $f(x|z)$

(b) $E(x|y)$

(c) $P[X < 3/4 | Y > 1/4]$.

3. Let $X_1, X_2, \dots, X_{1000}$ be a random sample of size 1000 from a normal distribution with mean zero and variance 1. Let $Y = \sum_{i=1}^{1000} X_i^2$. Find $P\{Y \geq 1005\}$.

4. Let X_1 and X_2 be independent normal random variables with mean θ . The variable X_1 has variance 1 and the variable X_2 has variance 2. Consider the following two tests for the hypothesis testing problem $H_0: \theta \leq 0$ vs. $H_1: \theta > 0$.

Test 1: Reject if $X_1 + X_2 > C_1$.

Test 2: Reject if $X_1 > C_2$ and $X_2 > C_3$, where C_1, C_2 , and C_3 are constants.

(a) Find C_1, C_2 , and C_3 so that the size of the tests at $\theta=0$ is .05.

(b) Find the power of each test when $\theta=1$.

5. (a) Find the maximum likelihood estimator for μ , the population mean, given a sample of size 2 from a population with density $f(x) = 1/\beta$; $0 < x < \beta$; $f(x) = 0$ otherwise. What is the distribution of the estimator? What are the mean and variance of the estimator?
- (b) Let $X_1, X_2, \dots, X_n$ be a random sample of size n from a uniform distribution with parameters μ and σ . That is

$$\begin{aligned} f(x; \mu, \sigma) &= 1/\sigma && \text{for } \mu - \sigma/2 < x < \mu + \sigma/2 \\ &= 0 && \text{otherwise.} \end{aligned}$$

Find the maximum likelihood estimates of μ and σ .

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Master's Comprehensive Examination

PART II

Applied

December 8, 1979

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1. Someone claimed that a person could not distinguish between two foods eaten when he could not smell, especially when the foods were about alike in texture and flavor. An expert selected two foods, A and B, as being types which would qualify under the claim if it were true. A test was arranged as follows:

Each of 100 persons was given four portions at random to taste; three of one type and one of the other. All wore nose clips during the test. Each was asked to pick out the "odd" portion.

- (a) State the null and alternative hypotheses.
- (b) Test the claim at the 5% level if 30 of the 100 people correctly identified the "odd" portion.
- (c) What conclusion can be drawn from this experiment?
2. An experiment was performed using eight tomato plants. One half of the flowers of each plant was pollinated by the experimenter using a new pollinating technique while on the other half of the plant the experimenter used a standard pollinating technique. The seed yield from each half of the plant is tabulated below.

<u>Plant No.</u>	<u>Seed Yield From Experimental Pollinating Technique</u>	<u>Seed Yield From Standard Pollinating Technique</u>
1	10	8
2	12	14
3	8	6
4	12	10
5	14	10
6	8	8
7	10	12
8	12	10

- (a) Determine if the experimental technique gives a higher yield in seed than the standard technique (let $\alpha = .05$). Be sure that both H_0 and H_a are shown.
- (b) How can the experimenter make a Type I error? What are the consequences of his doing so?

- (c) How can the experimenter make a Type II error? What are the consequences of his doing so?
 - (d) Give a 90% confidence interval for the difference between the two techniques.
3. A randomized block design was used as part of a field experiment to estimate the effects of eight fertilizers on the percent fiber in hay. From each plot six samples were selected and the fiber content of each was determined. The ANOVA conducted on the data was:

<u>Source</u>	<u>df</u>	<u>ms</u>
Total	239	
(F) Treatments	7	640
(R) Blocks	4	320
Experimental Error		160
Sampling Error		80

- (a) Fill in the missing df appropriate to the design and write out the ems.
 - (b) Define more specifically what the two sources of "Error" are in this particular experiment.
 - (c) Test the hypothesis that the population mean fiber content for the eight treatments are equal. Interpret your results.
 - (d) Assuming that the mean fiber content for one of the treatments is 20% compute and interpret its 95% confidence interval.
 - (e) Estimate the variance of a treatment mean if three blocks and four samples per plot were to be used. What would be the gain or loss in information compared to the use of four blocks and three samples per plot?
4. It is believed that the relationship between the % of catalyst in a chemical process and the yield is linear. From the data given below:
- (a) Fit the least squares regression line.
 - (b) Test for linearity assuming normal errors.
 - (c) Estimate the variance of the errors.
 - (d) Describe how one would proceed if the linear model is inadequate.

<u>% Catalyst (X)</u>	<u>Yield (Y)</u>
4	70
5	71
6	73
7	74
7	73
8	76
8	75
9	77
10	79
11	81

$$\Sigma X = 75 \quad \Sigma Y = 749 \quad \Sigma XY = 5684$$

$$\Sigma X^2 = 605 \quad \Sigma Y^2 = 56207$$

5. An experiment was conducted to compare the yield of different varieties of sweet potatoes. It was also desired to determine the effect of different concentrations of an insecticide on the yield. To answer these questions a factorial design was used to collect the data.

Complete the following ANOVA table.

<u>Source</u>	<u>df</u>	<u>ss</u>	<u>ms</u>	<u>ems</u>	<u>F</u>
Total	31	100			
(F) Variety					8
(F) Insecticide	1	4		$\sigma_P^2(VI) + 16\sigma_I^2$	
V x I	3				
(R) Plants(V x I)		24			

6. Four methods of performing a certain operation have been tried and ten observations for each method have been obtained. The mean productivity under each method is 3, 4, 6, and 7, respectively. While we do not have the original data we know that the coefficient of variation is 10%. Test the hypothesis that the four population means are equal.

RUTGERS-THE STATE UNIVERSITY

Master's Comprehensive Examination

Part I

Theory

March 10, 1979

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-

1. Suppose that a person is presented with one of two jars selected at random, and then draws two balls without replacement from that jar. If jar 1 has 5 black balls and 2 white balls and jar 2 has two black balls and 5 white balls, find the probability that

- a. the two balls come from jar 1, given that both are black
- b. the two balls come from jar 2, given that both are black

2. Given a sample of size n from a Poisson population with unknown mean λ , find the minimum variance unbiased estimator of $2\lambda^2 + \lambda$.
3. Based on a random sample of size n , for the density

$$f(x|\theta) = (1+\theta)x^\theta, \quad 0 \leq x \leq 1, \quad \theta > 0,$$

find the maximum likelihood estimator of

- a. θ
- b. $e^{2\theta} + \theta^2$

4. Based on a single observation from a density

$$f(x|\theta) = \frac{1}{\theta}, \quad 0 \leq x \leq \theta, \quad \theta > 0,$$

- a. find the Bayes estimator of θ under squared error loss with respect to prior density $g(\theta) = \theta e^{-\theta}$, $\theta > 0$.
- b. Find the mean and variance of the Bayes estimator in part a.

5. If x_1 and x_2 is a random sample of size two from a population with the uniform distribution over the unit interval, show that the distribution of $u = x_1 x_2$ is

$$f(u) = -\log u \quad 0 < u < 1.$$

6. Consider a hypothesis testing problem involving a normal distribution with mean μ and variance $\sigma^2 = 1$. We wish to test the null hypothesis $H_0: \mu = 0$ versus the alternative hypothesis $H_A: \mu = 1$. Based on a single observation x , we use the decision rule: Reject the null hypothesis if $-\frac{3}{4} < x < 0$, Accept the null hypothesis otherwise.

- a. What is α ?
- b. What is β ?
- c. What is the power of the test?
- d. Is this the "best" test that one can use for a critical region of this size?
- e. If the answer to (d) is yes, prove your statement. If it is no, find the "best" critical region and state in what sense it is best.

RUTGERS-THE STATE UNIVERSITY

Part II

Master's Comprehensive Examination

March 10, 1979

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1. Read this problem completely before you start its solution. A platelet is a cell component of blood that plays an important role in blood coagulation. The normal platelet count in humans is between 200,000 to 400,000 per cc of blood. Certain drugs used in treating cancer are known to lower the platelet count. Suppose that three different drugs D_1 , D_2 , D_3 are used at different dosage levels L_1 , L_2 , L_3 on four patients each. Suppose each patient has a platelet count of approximately 300,000 before treatment. We record this count as 300 (300×1000). After treatment, the counts for the 36 patients are taken. These counts are given below.

		<u>Drug</u>		
		D_1	D_2	D_3
<u>Dosage</u>	L_1	245	248	355
		302	213	298
		289	283	218
		252	255	315
	L_2	246	278	253
		250	194	263
		323	278	326
		235	298	234
	L_3	206	228	198
		254	241	190
		137	164	165
		237	191	231

Recopy the following summary table, filling in all entries marked by *. You may leave the F values as a ratio of two numbers. Show all calculations. (Hint: First fill in the df entries.)

<u>Source</u>	<u>df</u>	<u>SS</u>	<u>MS</u>	<u>F</u>
Drug (D)	*	1293.05	*	*
Level (L)	*	*	17240.30	*
(Drug)X(Level)	*	*	*	*
Within	*	*	1655.47	
Total	*	84010.97		

2. To study grade distributions we take a random sample of 300 students. The data are summarized as follows:

	<u>B or better</u>	<u>C or less</u>
Instructor I	25	35
Instructor II	55	95
Instructor III	20	70

- a. At the .02 level of significance, test

H_0 : Grade is independent of instructor

versus

H_1 : Grade is not independent of instructor

- b. Also, using the data from the above table as necessary, test the null hypothesis that:

25% of all students take Instructor I
 45% of all students take Instructor II
 30% of all students take Instructor III

Use $\alpha = .10$.

3. A researcher in physical education wishes to study whether a program of jogging will alter the resting pulse rate of men aged 40 to 50. Using a sample of 8 men, the results before and after completion of the program are:

<u>Subject</u>	<u>1</u>	<u>2</u>	<u>3</u>	<u>4</u>	<u>5</u>	<u>6</u>	<u>7</u>	<u>8</u>
Pulse rate (before):	74	86	80	98	85	83	74	92
Pulse rate (after):	70	85	82	90	82	79	71	89

- a. Why can't one use a 1 factor ANOVA to analyze this data?
 b. Perform the correct analysis of the data clearly stating the hypotheses, test statistic, decision rules and results of a 1% test.
 c. Give a 95% confidence interval for $\mu_{\text{before}} - \mu_{\text{after}}$.
4. The mean test score on a standardized psychological test given to a certain population is 550; the standard deviation is 20. Answer the following questions assuming the test scores are normally distributed. 243 individuals are randomly chosen from this population and tested.
- a. How many of the 243 individuals would you expect to have a score above 535?
 b. What is the probability that the 2nd individual tested scored above 569 and the 23rd individual tested scored below 542?
 c. What is the probability that the mean score of the first 25 individuals is between 542 and 547?

5. A psychologist seeks to evaluate the effectiveness of two methods of teaching reading. The psychologist sets up an experiment with 12 students of which six are boys and six are girls. The students within each sex are divided at random and assigned to the two methods of reading. The response variate is gain in reading ability. The results are as follows:

	<u>Boys</u>	<u>Girls</u>
Book Method of Teaching	1,2,3	4,4,4
Machine Method of Teaching	6,6,6	8,9,10

- Carry out the appropriate analysis of this data.
- Is there a significant difference between the different teaching methods?
- Does the difference between teaching methods differ for boys and girls?
- Suppose that we did not keep track of the fact that some of the students were boys and some were girls (but just assigned 6 students to the Book Method of Teaching and 6 to the Machine Method.), and suppose we got the above results. Namely,

Book Method: 1,2,3,4,4,4
Machine Method: 6,6,6,8,9,10

Carry out the appropriate ANOVA for this data.

- What are the disadvantages and advantages of the design in (d) as opposed to the originally given design?
6. An experiment was conducted to determine the relationship between the amount of training on a task and the performance on the task. Six individuals were given different amounts of training and then their performance was measured. The data was as follows:

<u>Individual</u>	<u>Performance on Task</u>	<u>Amount of Training (minutes)</u>
Jones	1	1
Smith	4	2
Davis	7	3
Brown	9	5
Green	9	9
Doe	10	7

- You seek to predict performance on a task for an individual with a given amount of training. Find the least-squares predicting equation.
- Use the predicting equation in (a) to estimate the performance for an individual with 4.5 minutes of training.
- Give a 99% confidence interval for the estimate in part (b).

RUTGERS-THE STATE UNIVERSITY

Master's Comprehensive Examination

Part I

Theory

December 9, 1978

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1. Let the joint density of (X, Y) be given by

$$f_{X,Y}(x,y) = A(x+2y) \quad \text{for } 1 < x < 2, 0 < y < 1, 1 < x+y < 2$$

$$0 \quad \text{otherwise}$$

- a. Find A .
 - b. Find $E(Y|X=x)$ for $x \in (1,2)$.
 - c. Find $E(XY|X=x)$ for $x \in (1,2)$.
2. You and I are each simultaneously flipping fair coins. We both stop when either you get your first head or you have flipped your coin three times.
 - a. Let N be the number of times you flip your coin and let X be the number of heads I will get. Find the joint distribution of N and X .
 - b. Find the expected value of X .
 - c. Find $\Pr(N=3|X=1)$.
 3. Let X be a sample of size 1 from a population with density $(\theta > 0)$

$$f_{X|\theta}(x|\theta) = \frac{1}{\theta}, \quad 0 < x < \theta.$$

It is desired to estimate θ with quadratic loss $L(a, \theta) = (a - \theta)^2$. Let the prior density of θ be given by

$$f_{\theta}(\theta) = \theta e^{-\theta}, \quad \theta > 0.$$

- a. Find the Bayes estimator of θ based on a sample of size one.
- b. Find the mean and variance of the Bayes estimator found in part (a).

4. Let X be a single observation from the density $f(x;\theta) = (1+\theta)x^\theta$, $0 < x < 1$ where $\theta > -1$.
- Find the most powerful size- α test of $H_0:\theta=0$ versus $H_1:\theta=1$. Give acceptance and rejection regions as functions of α .
 - Is there a uniformly most powerful size- α test of $H_0:\theta \leq 0$ versus $H_1:\theta > 0$? If so, what is it?
5. Let $X_1, \dots, X_n$ denote a random sample from

$$f(x;\theta) = 1 \quad \theta - \frac{1}{2} < x < \theta + \frac{1}{2} \quad \text{for} \quad -\infty < \theta < \infty.$$

Let $(\min(X_1, \dots, X_n), \max(X_1, \dots, X_n))$ be a confidence interval for θ .

- Show that the confidence coefficient for this interval does not depend upon θ .
 - Find the confidence coefficient of this interval.
6. Let $X_1, \dots, X_n$ be a random sample from a normal distribution with mean μ and variance one ($N(\mu, 1)$), where μ is an integer ($\mu = \dots, -1, 0, 1, \dots$).
- Find the maximum likelihood estimate of $\mu, \hat{\mu}$ (where $\hat{\mu}$ is an integer.)
 - Prove $E \hat{\mu} = \mu$.

RUTGERS-THE STATE UNIVERSITY

Master's Comprehensive Examination

Part II

Applied

December 9, 1978

- Instructions:
1. Open Book and Open Notes.
 2. Do four of the following six problems.
 3. Use a separate blue book for this section of the exam.
 4. Indicate which questions you wish graded.
-

1. Assuming X has a normal distribution how large a sample should be taken if we want to be 95% confident that $\bar{X}$ will not fall farther than $\sigma/2$ from μ ?
2. An entomologist wishes to conduct a study to compare four (4) insect repellents with regard to their effectiveness against black flies. Six individuals are available for the experiment. No more than two repellents may be used on an individual at a given time. That is, one may be placed on his left arm and one may be placed on his right arm. For purposes of this experiment it will be assumed that no difference exists between an individual's left and right arm. After the repellents are applied, all six (6) individuals are placed in a heavily infested area for a period of 15 hours. The response that is recorded is the number of black fly bites on each arm. The entomologist has only one day to conduct the experiment.
 - a) Give the name of the experimental design that you would recommend to the experimenter. Indicate why you feel the design is the best to use.
 - b) Indicate how the experimental treatments should be allocated and be specific.
 - c) Give the ANOVA table including the sources of variability, the degrees of freedom, and the expected mean square.
3. A new variety of tomato has been developed. It is desired to know if the yield of this variety is greater than the yield of a standard variety. An experiment was conducted and the yield was observed. The results are summarized below.

<u>New Variety</u>	<u>Standard Variety</u>
$n_N = 15$	$n_S = 10$
$\bar{X}_N = 125$	$\bar{X}_S = 100$
$s_N^2 = 85.7$	$s_S^2 = 94.4$

If a t-test is used the following is found:

$$t = \frac{\bar{X}_N - \bar{X}_S}{\sqrt{\frac{s_N^2}{n_N} + \frac{s_S^2}{n_S}}} = \frac{125-100}{\sqrt{\frac{89.1}{15} + \frac{89.1}{10}}} = 1.90$$

If an Analysis of Variance is used the following is found:

<u>Source</u>	<u>df</u>	<u>SS</u>	<u>MS</u>	<u>F</u>
Total	24	2361.8		
Variety	1	312.5	312.5	3.61
Error	23	2049.3	89.1	

- a) State the null and alternative hypothesis based on the experimenter's objective.
 - b) Can the t statistic be used to test the null hypothesis shown in part (a)? If yes, complete the test being sure to show the critical region. If no, explain why not.
 - c) Can the Analysis of Variance be used to test the null hypothesis shown in part (a)? If yes, complete the test being sure to show the critical region. If no, explain why not.
4. (a) A researcher wants to study the effectiveness of flu vaccination. He has asked each of 90 people whether he had had flu during the previous winter and also whether he had been vaccinated. Of the 90 people sampled, 18 had been vaccinated, and 45 had had flu. Of the 18 vaccinated people, only 3 had had flu. Should the experimenter conclude that vaccination and contracting flu are independent events or not?
- (b) A model for genetic inheritance predicts that 3 recognizable types (A,B, and C) should exist in a 6:1:1 ratio in a population. To test this model, 48 individuals were sampled at random and each classified according to type.

<u>TYPE</u>			
<u>A</u>	<u>B</u>	<u>C</u>	
30	6	12	48

Do the data suggest that the genetic model is a reasonable one for describing the inheritance of the trait being considered?

5. A simple linear regression model

$$Y = \alpha + \beta X + e$$

was fit to a set of data. The following results were found.

$b = 4$ where b is the least squares estimate of β

$$\bar{X} = 4$$

$$\bar{Y} = 20$$

Additionally part of the analysis of variance table is summarized below.

<u>Source</u>	<u>df</u>	<u>SS</u>	<u>MS</u>	<u>F</u>
Total	10	800		
Regression				
Error		288		

- Find the estimate regression line.
 - If it is assumed that e is $N(0, \sigma^2)$ then estimate σ^2 .
 - Test the hypothesis that $\beta = 0$. Let the probability of a Type I error = 0.10.
 - Find a 95% confidence interval for the mean value of Y when $X = 4$.
 - Find the coefficient of determination, r^2 . Interpret this statistic.
 - Estimate the correlation coefficient.
 - Test the hypothesis that $\rho = 0$.
6. A completely balanced nested experiment was used for comparing the means of ten (10) treatments. The following ANOVA resulted.

<u>Source</u>	<u>df</u>	<u>MS</u>	<u>ems</u>
(F) Treatments	9	570	
(R) Batches (TH)	90	190	
(R) Determinations (B(T))	200	10	

- Complete the ems column
- Are the 10 treatment means significantly different?
- Estimate the variance components for the random effects.
- Estimate the variance of a treatment mean.
- Assuming the components of variance do not change what would be the variance of a treatment mean if twenty (20) batches per treatment were selected and only a single determination was made from each batch?

RUTGERS-THE STATE UNIVERSITY

Master's Comprehensive Examination

PART II

Applied

December 10, 1977

- Instructions:
1. Open Book and Open Notes.
 2. Do 4 out of the following five problems.
 3. Use a separate blue book for this section of the exam.
 4. Indicate which questions you wish graded.
-

1. Samples of frozen food taken in New York in 1958, 1959, and 1960 showed the results in the table. Is there evidence of varying level over the years?

	1958	1959	1960
No. examined	240	278	144
No. satisfactory on several criteria	153	201	90

2. A marketing experiment was carried out by the importer of a foreign automobile to help determine a satisfactory pricing policy. A sample of 25 agencies was selected throughout the country. They were considered to be quite equal in sales potential. They were assigned certain sticker prices and were required to hold prescribed limits on them. Prices ranged from \$2500 to \$3000 per standard equipped car. After three months, the number of sales by agency and other statistical results were as follows:

Average price, $\bar{X} = \$2750$

Average sales, $\bar{Y} = 360$ per dealer

Estimate of the slope: $b = -.40$

Standard error of estimate: 20

$$\Sigma(X - \bar{X})^2 = 15625$$

- (a) Find the equation of the regression line.
- (b) Find the 95% confidence interval for the slope.
- (c) Find a 95% prediction interval for an individual forecast for Y given $X = \$2750$.
- (d) What change in sales does this experiment suggest would result if price of an automobile were reduced by \$100?

3. An important social topic is the effects of marijuana. Twenty-four college men are selected, 12 of whom are regular users of marijuana, ordinary cigarettes, and alcohol and 12 who are regular users of cigarettes and alcohol but who have never tried marijuana before. It is assumed that there are no other differences between these two groups of subjects (an assumption that may not be true). Within each group, user and non-user, six subjects smoked two marijuana cigarettes in the laboratory while six subjects smoked placebo cigarettes that taste and smell like marijuana but have no effect. A short time after smoking, all subjects took a set of perceptual-motor tests of skills required for driving. The researcher wanted to know if marijuana had more of an effect on driving skills than a placebo, if users responded differently from non-users, and if there was an interaction. The scores the subjects received are as follows:

Regular users receiving marijuana

42, 35, 48, 26, 55, 47

Regular users receiving placebo

35, 52, 43, 45, 49, 38

Naive users receiving marijuana

60, 45, 55, 39, 47, 42

Naive users receiving placebo

47, 42, 51, 37, 49, 48

Evaluate these data with the analysis of variance. Draw up a formal summary of the analysis, display all computation, graph any results and interpret the experiment.

4. Below is an abbreviated ANOVA table.

<u>Source</u>	<u>df</u>	<u>ms</u>
Total	31	-
A	3	240
B	3	120
AxB	9	60
obs(AxB)	16	30

In each of the following situations obtain the e.m.s., indicate the appropriate null and alternative hypotheses concerning A,B, and AxB, and test each hypothesis.

- (a) A is random and B is fixed.

5. Suppose we have a 2^4 experiment with factors A, B, C, D and wish to obtain a $\frac{1}{2}$ replicate by using ABD as the defining contrast.
- (a) What treatments in addition to (1) are used? [note: (1) is Yate's notation representing the treatment combination for which A, B, C, and D are all at their low levels.]
 - (b) How many experimental units are used?
 - (c) What are the aliases of all main effects and two factor interactions?
 - (d) Would you prefer another interaction in place of ABD in constructing a $\frac{1}{2}$ replicate? Why?

RUTGERS-THE STATE UNIVERSITY

Master's Comprehensive Examination

Part I

Theory

March 12, 1977

Instructions: Open Book and Open Notes. Do four of the following six problems. Use a separate blue book for this section of the exam. Be sure to indicate which questions you wish graded.

1. Someone rolls a pair of dice and tells you the result is odd and exceeds 6. What is the conditional probability that it is a 7 or 11?

2. Suppose n electron tubes are placed on life test. The experiment is stopped when all tubes have failed, the times of failure being $x_1, \dots, x_n$. Assume that the time-to-failure has an exponential distribution with probability density function given by

$$f(x; \theta) = \begin{cases} \frac{1}{\theta} e^{-\frac{x}{\theta}} & x \geq 0, \theta > 0 \\ 0 & x < 0 \end{cases}$$

where the parameter θ is the mean of the distribution.

- (a) Derive the maximum likelihood estimator $\hat{\theta}$ of the mean life θ . Show work in detail.

It can be shown that

$$\frac{2n\hat{\theta}}{\theta} = \chi_{2n}^2$$

That is $2n\hat{\theta}/\theta$ is chi-square with $2n$ degrees of freedom. This is the exact or small sample distribution for $\hat{\theta}$.

- (b) Show that $\hat{\theta}$ is an unbiased estimator for θ .
 - (c) Find $V(\hat{\theta})$, the variance of $\hat{\theta}$.
-
3. If x_1 and x_2 is a random sample of two from a population with the uniform distribution over the unit interval, show that the distribution of $u = x_1 x_2$ is

$$f(u) = -\log u \quad 0 < u < 1$$

4.
$$f_0(x) = (2\pi)^{-\frac{1}{2}} e^{-\frac{1}{2}x^2},$$

and

$$f_1(x) = (2\pi)^{-\frac{1}{2}} e^{-\frac{1}{2}(x-1)^2}.$$

Let $f = f_0$ be the null hypothesis and $f = f_1$ the alternate. A critical region,

$-\frac{1}{2} < x < 0$, is selected and a single observation taken.

- (a) What is the Type I error?
 - (b) What is the Type II error?
 - (c) What is the power of the test?
 - (d) Is this the "best" one can do for critical regions of this size?
 - (e) If the answer to (d) be "Yes" prove your statement. If the answer to (d) be "No" find the "best" region and state in what sense it is best.
5. Independent samples of sizes n_1 and n_2 , respectively, are drawn from two normal populations with equal means μ and with variances equal to σ^2 and $\lambda\sigma^2$.
- (a) Find the maximum likelihood estimate of μ if we use the information in both samples.
 - (b) Show that this estimate has variance

$$\frac{\sigma^2}{\frac{(n_1 + n_2)}{\lambda}}$$

6. X_1, X_2 are two independent observations from a uniform distribution in the interval $(0, \theta)$. Show that $(Y, \frac{Y}{\sqrt{\alpha}})$ is a confidence interval for θ with confidence coefficient $1 - \alpha$, where $Y = \max(X_1, X_2)$.

Statistical Methods
Master's Comprehensive Examination

Part II

Statistical Methods

March 12, 1977

Instructions: Open Book and Open Notes. Do four of the following six problems. Use a separate blue book for this section of the exam. Be sure to indicate which questions you wish graded.

1. From the simple linear regression model

$$Y = \beta_0 + \beta_1 X + \epsilon, \quad \epsilon \sim N(0, \sigma^2)$$

the following results were obtained

$$n = 11$$

$$\sum (X_i - \bar{X})^2 = 64$$

$$\bar{X} = 4$$

$$\sum (X_i - \bar{X})(Y_i - \bar{Y}) = 256$$

$$\bar{Y} = 20$$

$$\sum (Y_i - \bar{Y})^2 = 1600$$

- (a) Find the estimated regression line.
- (b) Find the minimum variance unbiased estimate of σ^2 .
- (c) Test the hypothesis that $\beta_1 = 0$.

Let the probability of a Type I error = .10.

- (d) Find a 95% confidence interval for a value of Y observed at some time in the future for $X = 4$.
- (e) Find an estimate of the correlation coefficient ρ .
- (f) Test the hypothesis that $\rho = .5$.

2. From some development data, it has been estimated that the Power Output on one radio component has an average, $\bar{P}_1 = 1500$ and $\sigma_1 = 30$; and a second component has $\bar{P}_2 = 2000$ and $\sigma_2 = 40$. The Power Output, $\bar{P}_g$, of the two components is additive in the set.

- (a) What prediction can you make about rejects in production if they produce components with the above averages and standard deviations and the upper specification on $\bar{P}_g$ is 3575 and the lower specification is 3400?
- (b) What confidence limits would you place on $\bar{P}_g$ if the initial averages and standard deviations were estimated from samples of size $n = 25$?
- (c) Would any other assumptions be helpful in supporting your predictions? (If "yes", please discuss.)

3. Upon calculating the analysis of variance of the pumping rates of oysters in six different water temperatures the corrected sum of squares for each of the four sources of variation (appropriate to the design) was found to be 245. The design was a randomized block in which eight different oysters were taken, two being assigned to each of the water temperature baths.

- (a) Make an abbreviated schematic of the design.
- (b) Complete the following analysis of variance showing the expected values of the mean squares.

Source of Variation	d.f.	Sum of Squares	M.S.	E.M.S.
Total				
Temperature				
Beds				
Temp. x Beds				
Oysters within Beds				

- (c) Test the significance of the temperature effect and make a probability statement of interpretation.
- (d) The two highest temperatures gave pumping rates of 41.7 and 44.7 (gallons per hour) -- the latter being for the highest temperature. At this range of temperatures has increasing the temperature significantly increased the pumping rate of the oyster? Support your answer with a probability statement.

4. A quality control engineer desires to characterize the variability in a process which is carried out by a large number of automatic machines running continuously. He suspects that the greatest variation is among machines but is also reasonably certain that the production from any one machine may not be consistently uniform. He knows nothing of the variability of the testing method but suspects it will be relatively small compared to product variation and is not willing to spend much effort to characterize it. He has two weeks in which to complete this study and can expend money for sixty (60) tests.

- (a) Propose a design which will best satisfy his needs. Show schematically.
- (b) Write out the analysis of variance for your design with the expected values of the mean squares using general notation if the design you propose is unbalanced.

5. Measurements of the yields of two processes, A and B, have shown average yields of $\bar{X}_A = 46.5$, $\bar{X}_B = 47.1$. The process yields are independent normal variables with means μ_A and μ_B and variances, $\sigma_A^2 = 25$ and $\sigma_B^2 = 16$ respectively. $\bar{X}_A$ is based on 100 observations and $\bar{X}_B$ on 64. We want to test $H_0: \mu_A = \mu_B$ with a 2-sided, 10% level test.
- Give the Critical Region of the test.
 - Carry out the test for the data above.
 - If $\mu_A = 50$ and $\mu_B = 48$, what is the power of this test?
 - Give a 90% confidence interval for $(\mu_A - \mu_B)$.
6. Assume that 2 Factors A and B have been included in an experiment, each factor at 2 levels. Construct a two way table (and a graph) using the numbers (24, 37, 57, 70) to illustrate each of the following hypothetical situations. (Note: you may use a number more than once in the same table).
- No A effect, Significant B effect, Zero interaction.
 - Significant A effect, Significant B effect, Zero interaction.
 - No A or B effects, Significant interaction.

RUTGERS-THE STATE UNIVERSITY

Master's Comprehensive Examination

Part I

Theory

December 18, 1976

- Instructions: 1. Open Book and Open Notes.
 2. Do four of the following six problems.
 3. Use a separate blue book for this section of the exam.
 4. Indicate which questions you wish graded.
-

1. Let $X_1, \dots, X_n$ be a random sample from a Poisson distribution with parameter λ .

- a) Find the MLE of $\lambda^2 + 5\lambda + 2$.
 b) Find a minimum variance unbiased estimator of $\lambda^2 + 5\lambda + 2$.

2. Let the joint density function of (X, Y) be given by

$$f_{X,Y}(x,y) = \begin{cases} 8xy & 0 < x < y < 1 \\ 0 & \text{otherwise} \end{cases}$$

- a) Find $E[Y|X = x]$
 b) Find $E[XY|X = x]$
 c) Find $\text{Var}[Y|X = x]$

3. Let X be a single observation on a population with density for $\theta > 0$ given by

$$f_{\theta}(X) = \begin{cases} \theta e^{-X\theta} & X > 0 \\ 0 & \text{otherwise} \end{cases}$$

- a) $(X, 2X)$ is a confidence interval for $1/\theta$. What is its confidence coefficient?
 b) Find another confidence interval with the same confidence coefficient but with smaller expected length.

4. A fair die is tossed until a 2 appears for the second time. Find the average number of tosses required.

5. A single observation is taken from one of the following distributions.

Values of X	0	1	2	3	4	5	6	7
$P_0\{X=x\}$	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{4}$
$P_1\{X=x\}$	$\frac{3}{32}$	$\frac{1}{16}$	$\frac{5}{32}$	$\frac{1}{8}$	$\frac{3}{32}$	$\frac{3}{16}$	$\frac{1}{8}$	$\frac{5}{32}$

Find the most powerful test of $H_0:P_0$ vs $H_1:P_1$ of size $\alpha = \frac{1}{2}$.

6. Let $X_1, \dots, X_n$ be Poisson with parameter λ . Let loss be squared error, and let the prior density of λ be given by

$$g(\lambda) = \frac{\lambda^{a-1} e^{-(\lambda/b)}}{\Gamma(a)b^a}$$

- a) Find the Bayes estimator of λ .
b) Find the bias of the estimator found in part a).

$$(\text{Bias } (\delta) = E_{\lambda}(\delta - \lambda)).$$

RUTGERS-THE STATE UNIVERSITY

Master's Comprehensive Examination

Part II

Applied

December 18, 1976

Instructions: 1. Open Book and Open Notes.
 2. Do four of the following six problems.
 3. Use a separate blue book for this section of the exam.
 4. Indicate which questions you wish graded. - - - - -

1. Assume that 2 factors A and B have been included in an experiment, each factor at 2 levels. Construct a two way table (and a graph) using the numbers (24, 37, 57, 70) to illustrate each of the following hypothetical situations. (Note: you may use a number more than once in the same table).

- (a) No A effect, Significant B effect, Zero interaction
- (b) Significant A effect, Significant B effect, Zero interaction
- (c) No A or B effects, Significant interaction.

2. From the simple linear regression model

$$Y = \beta_0 + \beta_1 X + \epsilon \quad \epsilon \sim N(0, \sigma^2)$$

the following results were obtained

$$n = 11$$

$$\sum (X_i - \bar{X})^2 = 64$$

$$\bar{X} = 4$$

$$\sum (X_i - \bar{X})(Y_i - \bar{Y}) = 256$$

$$\bar{Y} = 20$$

$$\sum (Y_i - \bar{Y})^2 = 1600$$

- (a) Find the estimated regression line.
- (b) Find the minimum variance unbiased estimate of σ^2 .
- (c) Test the hypothesis that $\beta_1 = 0$.

Let the probability of a Type I error = .10.

- (d) Find a 95% confidence interval for a value of Y observed at some time in the future for $X = 4$.
- (e) Find an estimate of the correlation coefficient ρ .
- (f) Test the hypothesis that $\rho = .5$.

3. For each of the following situations, assume that an Analysis of Variance is to be performed. List the sources of variation, the degrees of freedom and the expected mean square. Be sure to indicate which sources of variation are fixed and which are random.

- (a) An experimenter wants to compare the percent acid content of 3 varieties of tomatoes. Each of 6 state experiment stations has agreed to grow the 3 varieties. Each station will homogenize a randomly selected tomato of each variety, determine the percent acid content, and submit this one measurement (for each variety) to the experimenter.

- (b) An experimenter wants to compare the percent acid content of 3 varieties of tomatoes. From a single experiment station he obtains a bushel of each variety. For each variety he then selects 6 tomatoes at random from the bushel and records the percent acid content of each of the selected tomatoes.
- (c) An experimenter wants to compare the percent acid content of 3 varieties of tomatoes. In a greenhouse, 4 benches were used. Placed on each bench was a single plant from each variety. From each plant 2 fruit were selected and the percent acid content of each fruit was recorded.
- (d) An experimenter wants to compare the percent acid content of 3 varieties of tomatoes. He finds 6 home gardeners who are growing variety A, 6 who are growing variety B, and 6 who are growing variety C. Each gardener will supply the experimenter with 2 tomatoes and the experimenter will then record a measurement of percent acid content for each tomato.

4. Indicate which of the following statements are true (T) and which are false (F).

- (a) The Randomized Complete Block Design (RCBD) seeks to group experimental units into groups where the units within each group are homogeneous in some "important" sense.
- (b) The RCBD works best when the blocks are similar to each other.
- (c) In a RCBD the blocks are usually of little interest in themselves, but have been included in the design to "improve" our ability to compare treatments.
- (d) In the Analysis of Covariance the F-statistic for comparing adjusted treatment means will always be larger than the F-statistic for comparing unadjusted treatment means.
- (e) The Analysis of Covariance attempts to see how the treatments would have compared if they had been tested under the same conditions with respect to the covariate.
- (f) Analysis of Covariance has an advantage over blocking in that sometimes one can remove the effects of a source of extraneous variation even though that source was not recognized until after the experiment was underway or even completed (too late to block).
- (g) Treatment means can be adjusted for the effects of covariates in designs more complicated than the Completely Randomized Design.
- (h) Making α smaller actually increases the probability of making a Type II error if H_0 is false.
- (i) If the sample variances of the different groups vary greatly then there is reason to believe that the F test associated with the Analysis of Variance may not be applicable.

- (j) The smaller the value of α , the more probability there is that H_0 will be accepted.

5. A nutritionist is interested in comparing the weight gains of guinea pigs fed on four different diets. For each diet 4 animals were selected at random from a group of 16 animals. The diets were then administered for a specified period of time, after which the weight gains were recorded.

	<u>Animals</u>					$\sum_j X_{ij}$	$\sum_j X_{ij}^2$
	1	2	3	4	5		
Diet	1	2	0	-2	0	0	8
	2	0	2	0	-2	0	8
	3	5	5	2	8	20	118
	4	4	0	5	3	12	50
$\sum_i X_{ij}$	11	7	5	9	$\sum_{i,j} X_{ij} = 32$		
$\sum_i X_{ij}^2$	45	29	33	77	$\sum_{i,j} X_{ij}^2 = 184$		

- (a) Test the hypothesis that the mean weight gain is the same for all 4 diets.
- (b) Obtain a 95% confidence interval for $\left[\frac{\mu_1 + \mu_2}{2} - \frac{\mu_3 + \mu_4}{2} \right]$
- (c) Using a multiple comparison test of your choice establish which means are significantly different.
6. A friend claims to have E.S.P. (extra-sensory perception). He states that he can predict the result of a toss of a coin with more frequent success than chance would allow.

We believe he is only guessing and decide to test him with 15 flips of a fair coin. Before each flip he "predicts" the result, then we flip the coin and record whether or not he was correct. Assume the trials are independent and identical.

The following questions may be answered using a normal approximation.

- (a) How many correct predictions must he make in 15 trials to persuade us ($\alpha = .05$) that he has these remarkable powers?

H_0 : He is guessing

H_a : He has E.S.P.

- (b) If in fact our friend actually can correctly predict the outcome with probability .80, what is the probability that the test you just constructed in part (a) will result in a Type II error?
- (c) Suppose our friend makes 35 correct predictions in 50 trials. Obtain a 95% confidence interval for the true probability that he correctly predicts the outcome of a flip of a coin.

RUTGERS-THE STATE UNIVERSITY

Master's Comprehensive Examination

Part I

Theory

March 13, 1976

- Instructions:
1. Open Book and Open Notes.
 2. Do four of the following six problems.
 3. Use a separate blue book for this section of the exam.
 4. Indicate which questions you wish graded.
-

1. Someone rolls a pair of dice and tells you the result is odd and exceeds 6. What is the conditional probability that it is a 7 or 11?
2. Let X and Y be random variables with a joint density given by

$$f(x,y) = \begin{cases} 2 & 0 < x < y < 1 \\ 0 & \text{otherwise} \end{cases}$$

Find (a) The marginal density of Y

and (b) $P[Y < 2X]$

(c) Are X and Y independent? Prove or Disprove.

3. Given a sample of size n from a Poisson population with unknown mean λ , find the minimum variance unbiased estimator of $\lambda^2 + 3\lambda$.
4. Let X be a single observation on a population with density

$$f_{\theta}(x) = \begin{cases} \frac{3x^2}{\theta^3} & 0 < x < \theta \\ 0 & \text{otherwise} \end{cases}$$

Find a $100(1-\alpha)\%$ confidence interval for θ .

5. Let $X_1, \dots, X_n$ be a random sample from a population with density given by

$$f_{\lambda}(x) = \begin{cases} \lambda e^{-\lambda x} & x > 0 \\ 0 & \text{otherwise} \end{cases}$$

(a) Find the M.L.E. of λ .

(b) Find the Bayes estimator of λ if loss is squared error and the prior has density $g(\lambda) = e^{-\lambda}$, $\lambda > 0$.

6. Let $f_0(x) = \frac{e^{-\frac{1}{2}x^2}}{\sqrt{2\pi}}$ and $f_1(x) = \frac{e^{-\frac{1}{2}(x-1)^2}}{\sqrt{2\pi}}$.

It is desired to test $H_0: f=f_0$ vs $H_1: f=f_1$. A single observation X is taken

and a critical region $0 < x < 1$ is selected. For this critical region,

- (a) Find α .
- (b) Find β , the power.
- (c) Is this the best critical region of its size? If not what region is best?

RODGERS-LEE STATE UNIVERSITY

Master's Comprehensive Examination

Part II

Statistical Methods

March 13, 1976

Instructions: Open Book and Open Notes. Do four of the following six problems. Use a separate blue book for this section of the exam. Be sure to indicate which questions you wish graded.

1. A new packaging material was compared against a standard material. The test was conducted by storing the same food product in each of these for a fixed period of time and then determining the moisture loss. Each material was used to wrap ten (10) food samples. The following results were obtained.

	<u>New Material</u>	<u>Old Material</u>
Sample size	10	10
Mean moisture loss	224	232
$\sum(x_i - \bar{x})^2$	300	400

- (a) Conduct a statistical test to determine if the new material has a lower mean moisture loss than the old material. Let $\alpha = .05$.
- (b) Find a 95% confidence interval for the difference between the two mean moisture losses.
2. (a) Compute Y 's regression on X using the following data and the model $E(Y) = \beta_0 + \beta_1 X$.

Y	2	3	4	4	5	7	7
X	-3	-2	-1	0	1	2	3

- (b) Is the slope significantly different from zero?
- (c) Find the 95 percent confidence interval for the $E(Y)$ when $X=1$.
3. In order to determine the effect of dietary iron sources on the hemoglobin content of the blood of mice, four experimental diets were fed to weanling mice, with ten (10) mice assigned at random to each diet. After an appropriate time on the diets a sample of blood was taken from each mouse and the forty (40) samples were subjected to duplicate analyses for hemoglobin.
- (a) Make a schematic representation of this experimental plan.

- (b) Complete the following analysis of variance and write out the expected values of the mean squares:

<u>Source of Variation</u>	<u>d.f.</u>	<u>Sum of Squares</u>	<u>M.S.</u>	<u>E.M.S.</u>
Total	79	6360		
Diets		1800		
Mice within Diets			100	
Analyses within Mice within Diets				

- (c) Test the significance of the effect of the diets on the hemoglobin content of the blood and make a probability statement of interpretation.
- (d) Which is the greater source of experimental variation — Mice or Analyses? Is the difference in variability due to these sources statistically significant? Support your conclusion with a probability statement.
- (e) What magnitude of difference in the hemoglobin content of two of the dietary means would be needed to be considered "significant at the 1% probability level"?
- (f) What is the standard deviation due to analysis?

4. A small bottle (sample) from each of 25 consecutive production batches was analyzed by each of two analysts, A and B.

Present evidence pertaining to such important matters as:	<u>Batch No.</u>	<u>Analyst A</u>	<u>Analyst B</u>
	1	93	93
(a) stability of the production process, and	2	89	91
	3	84	93
	4	90	90
(b) agreement between the two analysts.	5	87	87
	6	90	90
(c) Discuss implications.	7	93	92
	8	89	89
	9	89	90
	10	90	90
	11	90	90
	12	92	89
	13	92	88
	14	92	92
	15	92	93
	16	90	90
	17	90	90
	18	90	90
	19	87	92
	20	92	93
	21	88	88
	22	89	89
	23	94	94
	24	93	93
	25	91	92

5. An experiment was conducted to determine the effect of temperature on the rate of reaction of a chemical experiment. The temperatures considered were 200 degrees C., 250 degrees C., and 300 degrees C. At each temperature four (4) experiments were run and the rate of reaction was recorded. An ANOVA was performed on this data and it was found that $\text{Error}_{ms} = 9.0$. It was also found that the average rate of reaction for each temperature was as follows.

$$T_{200} = 1$$

$$T_{250} = 4$$

$$T_{300} = 7$$

- (a) Completely write out the ANOVA table.
- (b) Perform the appropriate F test, state the null and alternative hypothesis, and state the conclusions you draw from this test.
- (c) Use the l.s.d. to compare the three treatments. What conclusions do you draw from this test?

6. From the production line for the manufacture of glass bottles a sampling study was conducted to investigate the number of defective bottles. Samples of 100 bottles were chosen at random from each of three machines on each of the three shifts on five consecutive production days.

The numbers of defective bottles/100 bottles inspected were subjected to analysis of variance appropriate to the design of the study. The following is a partial summary of the ANOVA.

<u>Source</u>	<u>df</u>	<u>Sums of Squares</u>	<u>Mean Squares</u>	<u>E.M.S.</u>
Total	44	1179		
Machines			115.50	
Dates		259		
MxD			12.25	
Shifts		156		
MxS			2.25	
DxS		150		
MxDxS				

- (a) Complete the ANOVA shown above assuming Machines and Dates to be Random and Shifts to be Fixed effects.
- (b) Make a complete statistical interpretation of this analysis using the following means and some sort of appropriate intervals.

Machine 1	8.2	Shift A	7.9	Day 1	7.7
2	9.5	B	10.9	2	7.7
3	13.5	C	12.4	3	10.7
				4	13.7
				5	12.3

- (c) In an effort to reduce variability in defects what steps would you propose?
- (d) Assuming that Machine, Date and Shift variation could be eliminated to that which might be due to random "uncontrollable" variation what is your best estimate of the standard deviation of such an improved process?